ABSTRACT

JavaLabPortal — Java Practical Lab Learning & Examination Portal is a centralized web-based educational application designed to modernize traditional, manual engineering practical laboratory record keeping and examination for **Rajasthan Technical University (RTU)** 5th Semester students. It provides an interactive digital representation of the institutional computing laboratory, replacing fragmented paper-based records with structured, accessible online syllabus workflows. The system establishes dedicated, role-differentiated dashboards for **Teachers** and **Students**, enabling dynamic code visualization, live arithmetic calculation, and continuous instructor feedback through an accessible, responsive user interface.

The application is built using **Java 17 (JDK 17)**, **Jakarta/Java Servlet 4.0**, **JavaServer Pages (JSP 2.3)**, **Apache Tomcat 9**, **MySQL 8.0**, **HTML5**, **CSS3**, and **JavaScript**. Its core architectural features include centralized HTTP security interception via **AccessControlFilter**, strict **Role-Based Access Control (RBAC)**, academic class section and batch allocation, live student roster management (CRUD), and dynamic mathematical expression evaluation using the **Shunting-Yard algorithm** with an operator precedence stack. The portal provides comprehensive unit-wise practical experiment coverage (Experiments 1 to 10), **PreparedStatement** sanitization to prevent SQL injection, Cross-Site Scripting (XSS) defenses, and containerized cloud deployment on **Docker** and **Railway**.

JavaLabPortal is an **academic institutional prototype** engineered to bridge classroom theoretical instruction with practical laboratory implementation. The project demonstrates the disciplined application of the standard **3-tier Model-View-Controller (MVC)** design pattern, Third Normal Form (**3NF**) relational database normalization, and stateful session tracking. By unifying syllabus experiment demonstrations, interactive calculators, faculty study material, and student evaluation into a singular paperless platform, JavaLabPortal establishes a reliable foundation for coursework evaluation and practical examination.